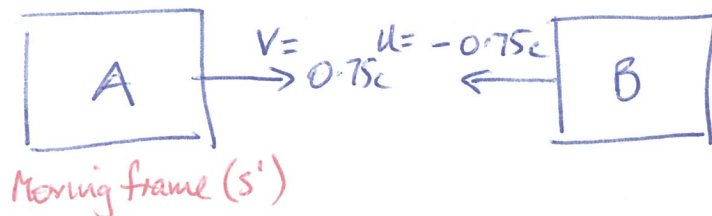


ME
(S frame)
Stationary.



I observe: $A = +0.75c$ relative to me
 $B = -0.75c$ relative to me. } Can be considered relative to Earth.

My frame is stationary. (S frame)

Consider B as an event - like a missile or object moving relative to A.

$\Rightarrow$ Take A as the moving reference frame S'

$\Rightarrow$ Its velocity is V .

$\therefore$ The velocity of B is $u = -0.75c$

QUESTION What is the velocity of B relative to A?
 i.e. what is u' ?

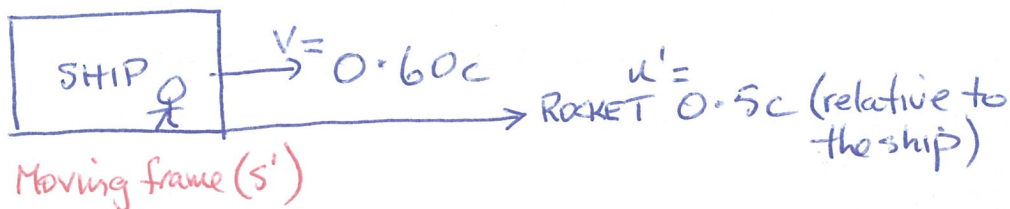
$$\begin{aligned} u' &= \frac{u - V}{1 - \frac{V u}{c^2}} \\ &= \frac{-0.75c - 0.75c}{1 - \frac{(0.75c)(-0.75c)}{c^2}} \\ &= \frac{-1.50c}{1.56} \\ &= \underline{-0.961c} \end{aligned}$$

which is consistent with the sign convention.

ME
(s frame)



STATIONARY



I observe: The ship is moving away at $0.60c$.
 $\Rightarrow v = 0.60c$

Observer on ship: sees rocket move at $0.5c$
 $\Rightarrow u' = 0.5c$

QUESTION: What do I see the rocket move at in my frame?
i.e. what is u ?

$$u = \frac{v + u'}{1 + \frac{vu'}{c^2}}$$

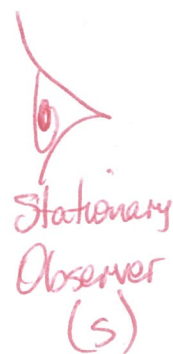
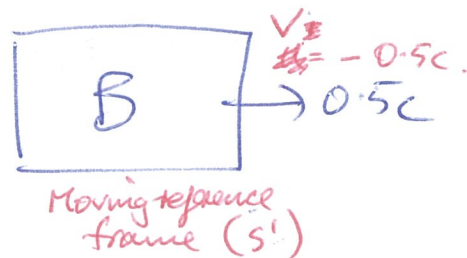
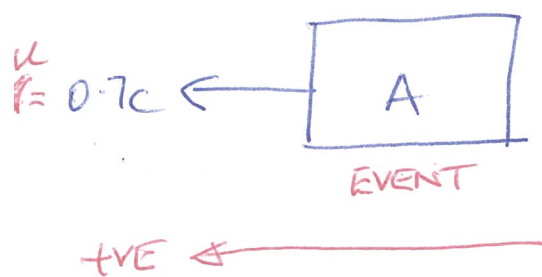
$$= \frac{0.60c + 0.5c}{1 + \frac{(0.60c)(0.5c)}{c^2}}$$

$$= \frac{1.10c}{1.30c}$$

$$= \underline{0.846c}$$

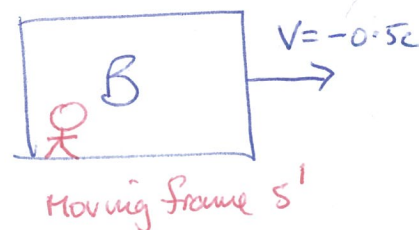
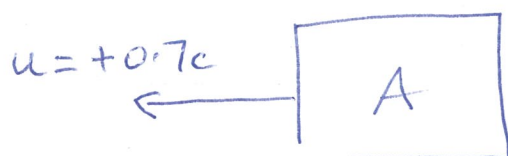
Sign is consistent with the convention.

Find V_{AB} - velocity of A relative to B.



$$\begin{aligned}
 u' &= \frac{u - v}{1 - \frac{vu}{c^2}} \\
 &= \frac{0.7 - (-0.5c)}{1 - \frac{(-0.5c)(0.7c)}{c^2}} \\
 &= \frac{1.2c}{1.35c} \\
 &= 0.889c
 \end{aligned}$$

A is moving $0.889c$ away from B.



$$\begin{aligned}
 u' &= \frac{u - v}{1 - \frac{vu}{c^2}} \\
 &= \frac{0.7c - (-0.5c)}{1 - \frac{(-0.5c)(0.7c)}{c^2}} \\
 &= \frac{1.2c}{1.35c} \\
 &= \underline{0.889c \text{ away from B (+ve)}}
 \end{aligned}$$